Case 12: WUI Temporal-Resolution Sensitivity

1 Verification purpose

This case verifies temporal-resolution sensitivity using a fixed physical problem and four level-set timesteps. Every curve identified as ELMFIRE output is regenerated from case-specific simulation products.

This case verifies temporal-resolution sensitivity in a firebrand-driven WUI calculation. Spatial resolution, geometry, wind, heat-release history, firebrand physics, random seed, and output cadence are fixed; only the level-set timestep changes over a wide Courant–Friedrichs–Lewy (CFL) range.

The test determines the range over which ignition chronology and structure-level spread remain timestep independent and makes loss of resolution explicit at very large steps. Success is evaluated against the finest-step baseline using load, ignition-time, and spread metrics.

2 Mathematical formulation and numerical implementation

2.1 Fixed physical problem

The community is a 220×100-m two-dimensional lattice of 10 m square structures separated by 10 m inert firebreaks in both directions. Every structure in the first upwind column is burning at $t = 0$, and wind is uniform at 40 mph from west to east. Non-structure cells use nonburnable fuel model 93, so vegetative surface-fire spread is absent. Thus a new source can appear only after deposited firebrands complete the configured physical ignition process.

For a structure ignited at t_i, the design heat-release rate (HRR) per unit area is

$$q''(\tau) = \begin{cases} q''_{\max}\tau/300, & 0 < \tau < 300, \\ q''_{\max}, & 300 \le \tau \le 3900, \\ q''_{\max}(4200-\tau)/300, & 3900 < \tau < 4200, \\ 0, & \text{otherwise}, \end{cases} \qquad \tau = t - t_i, \qquad q''_{\max} = 400\,\mathrm{kW\,m^{-2}}.$$

For footprint area $A_s = 100\,\mathrm{m}^2$, $\mathrm{HRR}_i = q''A_s$. Structural firebrands are emitted using the per-power rate

$$\dot{N}_i(t) = GR' \times \mathrm{HRR}_i(t), \qquad GR' = 10\ \mathrm{pcs\,MW^{-1}\,s^{-1}}.$$

2.2 Transport, deposition, and analytical load

The downwind landing distance X is represented by the lognormal density

$$f(x \mid \mu, \sigma) = \frac{\exp[-(\ln x - \mu)^2/(2\sigma^2)]}{x\sigma\sqrt{2\pi}}, \qquad x > 0.$$

For source structure i, the heat-release-dependent transport parameter is

$$B_i^* = \frac{U}{\sqrt{gL}} \left(\frac{\rho_f}{\rho_a}\right)^{-3/4} \left(\frac{d_f}{L}\right)^{-3/4} \left[\frac{\mathrm{HRR}_i}{\rho_a c_p T_a \sqrt{g}\, L^{5/2}}\right]^{1/2}.$$

The dimensional mean and standard deviation of landing distance are

$$\bar{x}_i = 0.47(B_i^*)^{2/3}L, \qquad s_{x,i} = 0.88(B_i^*)^{1/3}L,$$

and the log-space parameters supplied to f are

$$\sigma_i = \sqrt{\ln[1 + (s_{x,i}/\bar{x}_i)^2]}, \qquad \mu_i = \ln\left(\frac{\bar{x}_i}{\sqrt{1 + (s_{x,i}/\bar{x}_i)^2}}\right).$$

Here $L = 10\,\mathrm{m}$, $d_f = 5 \times 10^{-3}\,\mathrm{m}$, $\rho_f = 100\,\mathrm{kg\,m^{-3}}$, $\rho_a = 1.1\,\mathrm{kg\,m^{-3}}$, $c_p = 1\,\mathrm{kJ\,kg^{-1}\,K^{-1}}$, $T_a = 300\,\mathrm{K}$, $g = 9.81\,\mathrm{m\,s^{-2}}$, and $U = 0.447(40)/0.87 = 20.55\,\mathrm{m\,s^{-1}}$. This parameterization and its configured constants follow the documented structural-firebrand model [1]; the equations are reproduced here so the reference can be evaluated without consulting that source.

The independently calculated deposited count on target structure k is

$$N_k = \sum_{i<k} \int_0^T GR' \,\mathrm{HRR}_i(t) \left[\int_{x_k}^{x_{k+1}} f(x - x_i \mid \mu_i(t), \sigma_i(t))\, dx\right] dt.$$

The inner integral is evaluated using the lognormal cumulative distribution function (CDF) and the time integral uses an independent 5 s midpoint quadrature. ELMFIRE's final deposited count is divided by the physical $100\,\mathrm{m^2}$ structure footprint, giving N_k/A_s in $\mathrm{pcs\,m^{-2}}$.

The Eulerian firebrand tracker advances with its own wind-limited substeps. Changing the level-set step therefore must not change the spatial resolution of the flight trajectory. This is the core algorithmic separation being tested. In the generated namelists, prescribed timestep and maximum timestep impose the four values above. The input target surface-spread Courant number remains the ordinary surface-spread limiter and does not redefine the four wind-based temporal simulation conditions. The global vegetative surface-spread setting switch remains true to suppress vegetative surface-fire propagation. The current source independently refreshes initialized model-14 building HRR, and the nonburnable fuel-93 background provides an additional isolation of building-only spread. The fractional timesteps are intrinsic to this temporal-resolution sweep. For each member, preprocessing sets the generated stop to $\lceil t_{\text{requested}}/\Delta t \rceil \Delta t$, and records the requested stop, aligned stop, and integer step count in the simulation specification.

2.3 Physical ignition

For the configured structural ignition model, the critical deposited load is $N_{\text{crit}} = 2928.9\,\mathrm{pcs\,m^{-2}}$, the smoldering-to-flaming timescale is 42.1 s, and the subsequent structure development delay is 300 s [1]. Because these are much longer than the CFL=0.5, 1, and 10 level-set steps, their ignition chronologies should remain nearly unchanged. CFL=200 deliberately makes the time step comparable to or larger than the short ignition time scale and exposes the onset of temporal sensitivity.

3 Derivation of expected behavior

When firebrand flight and ignition time scales are much longer than the level-set step, moderate timestep changes should preserve deposited-load histories and structure ignition times. Extremely large steps should show increasing temporal error without changing the fixed physical problem.

At peak HRR, one 10×10 m structure produces

$$HRR_{\text{max}} = 400(100) = 40\ \mathrm{MW}, \qquad \dot{N}_{\text{max}} = 10(40) = 400\ \mathrm{pcs\,s^{-1}},$$

and the critical total load is $2928.9(100) = 292890$ particles. The transport reference uses the adjusted air speed $0.447(40)/0.87 = 20.55\,\mathrm{m\,s^{-1}}$, whereas preprocessing constructs the declared level-set

timesteps from the 20-ft input speed $U_{20} = 40(0.44704) = 17.8816\ \mathrm{m\,s^{-1}}$. Thus

CFL_w	0.5	1	10	200
$\Delta t = CFL_w(10)/17.8816$ (s)	0.279617	0.559234	5.592341	111.846815

which reproduces the simulation condition names to their displayed precision.

Relative to the 42.1 s small-fuel timescale, these steps are $0.00664,\ 0.0133,\ 0.1328$, and 2.657 times as large. The first three therefore resolve that timescale, while CFL 200 spans more than twice it and is expected to show temporal degradation. The analytical convolution is unchanged among simulation conditions, so its ideal load error is zero. The explicit numerical substitutions are: baseline mean load error at most 0.01; CFL 1 and 10 maximum load and mean ignition-time errors at most 0.001; their history NRMSE at most 0.05; and CFL 200 maximum load error at most 0.10.

4 Verification metrics and acceptance criteria

Prior calculations for this fixed physical problem motivate the following predeclared expectations [1]:

- the maximum analytical-load error remains below 10% even at CFL=200;
- for CFL no greater than 10, analytical-load errors are generally below 0.1%;
- ignition-time errors relative to CFL=0.5 remain below 0.1% for CFL no greater than 10;
- accumulation histories at the structure spanning $140 \le x \le 150$ m closely align; and
- first-deposition timing remains close through the approximately 80 m peak-HRR spotting range.

The case converts those expectations into explicit acceptance checks. The CFL=0.5 mean error against the analytical deposited-load integral defined above must not exceed 1%; the CFL=1 and 10 maximum errors against the same integral must not exceed 0.1%; and the CFL=200 maximum must remain below 10%. CFL=1 and 10 mean full-ignition-time errors relative to CFL=0.5 must not exceed 0.1%. Accumulation-history normalized RMSE at 140–150 m must not exceed 5% for CFL=1 and 10. Every simulation condition must be complete, and every acceptance component must pass.

The postprocessor also estimates first-deposition time from the first transient output interval with nonzero accumulation. Because its resolution is the 500 s dump interval rather than the firebrand trajectory substep, this is shown as a diagnostic corresponding to the first-deposition diagnostic, but is not used as a pass/fail metric.

4.1 Justification of metric quantification

The 0.1% limits for CFL 1 and 10 and the 10% limit for CFL 200 quantify the reported temporal-resolution behavior for this fixed-grid experiment rather than being selected from the present outputs [1]. They deliberately ask for near invariance over the moderate-step range while treating the extreme-step case as acceptable only if its error remains bounded rather than converged. The baseline mean-load limit is 1% because it compares a cell-integrated analytical deposition relation with a thresholded numerical field; it establishes that the reference member itself is accurate enough to anchor the relative ignition comparisons.

The 5% history NRMSE limit is wider than the final-load limits because transient rasters are sampled every 500 s; small timing offsets at a rising accumulation front are amplified pointwise even when final totals agree. Normalizing by the baseline history range makes 5% a direct bound on the fraction of the

resolved accumulation excursion. First-deposition time is kept diagnostic because its 500 s observation interval is too coarse to support a defensible tighter decision. All four simulation conditions are required so the case demonstrates both moderate-step invariance and the predicted extreme-CFL degradation.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	Current-input identity record outputs for baseline and CFL 1, 10, and 200 simulation conditions.	4 of 4 simulation conditions
Baseline load mean error	Mean structure-load relative error against the analytical load reference.	≤ 0.01
CFL 1 load maximum error	Maximum finite structure-load relative error against its analytical reference.	≤ 0.001
CFL 10 load maximum error	Same maximum error at CFL 10.	≤ 0.001
CFL 200 load maximum error	Same maximum error at CFL 200.	≤ 0.10
CFL 1 ignition error	Mean matched full-ignition-time relative error against CFL 0.5.	≤ 0.001
CFL 10 ignition error	Mean matched full-ignition-time relative error against CFL 0.5.	≤ 0.001
CFL 1 history NRMSE	Normalized RMSE of accumulation history at the 140–150-m structure against baseline.	≤ 0.05
CFL 10 history NRMSE	Same normalized history error at CFL 10.	≤ 0.05
Overall decision	Conjunction of completeness and all eight quantitative checks.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for the wind Courant number 0.5 and timestep 0.28 s condition. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

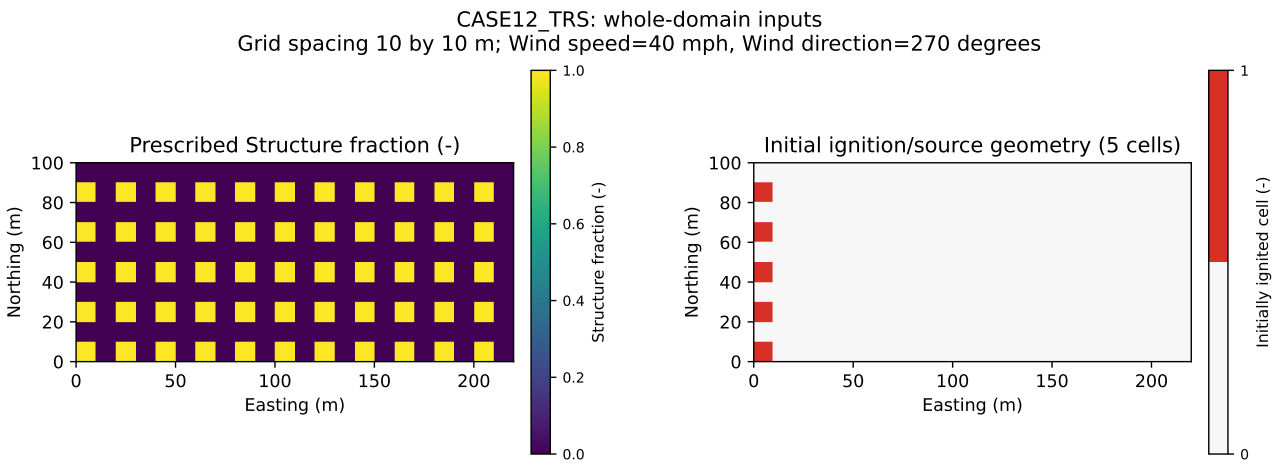


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Table 2: Generated temporal-resolution simulation conditions. The time step is calculated from $\Delta t = \mathrm{CFL}\,\Delta x/U$, not rounded to the display value.

Simulation condition	Wind-based CFL	Δt (s)	$t_{\max}$ (s)
Wind Courant number 0.5	0.5	0.27962	22000
Wind Courant number 1	1.0	0.55923	22000
Wind Courant number 10	10.0	5.5923	22000
Wind Courant number 200	200.0	111.85	22034

All simulation conditions use $\Delta x = \Delta y = 10\,\mathrm{m}$, alternating 10 m structures and firebreaks, a 40 mph wind, the same model-14 design fire, per-MW firebrand generation, empirical structural transport, Eulerian accumulation, physical ignition, no ember consumption, no crosswind distribution, and no vegetative surface-fire pathway. The predicted result is near-overlap for CFL=0.5, 1, and 10, with visible but bounded load sensitivity at CFL=200.

5.1 Preprocessing, execution, and postprocessing

Each prescribed condition uses co-registered fields in a Universal Transverse Mercator coordinate system and the stated fuel properties. The initial level-set field ϕ is negative within every structure footprint in the first upwind column. The exact timestep and input identities are recorded for each condition. Unique structure identities preserve the two-dimensional lattice; the profiles compare a representative cross-stream row.

A completed result is reused only when its input identity matches the current prescribed configuration. Each simulation condition is evaluated independently.

result analysis procedure aggregates final ember counts and ignition times by physical structure ID, reconstructs cumulative histories from transient ember rasters and recorded output times metadata, evaluates the analytical deposited-load relation, calculates every declared metric, and produces report-ready vector figures. Stale pre-modification outputs are explicitly excluded.

6 Actual simulation results

Figure 2 shows the representative time-of-arrival field from the wind Courant number 0.5 and timestep 0.28 s condition. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. The finest-timestep reference field anchors the profile and error comparisons and exposes domain-wide behavior that a single-row profile can hide.

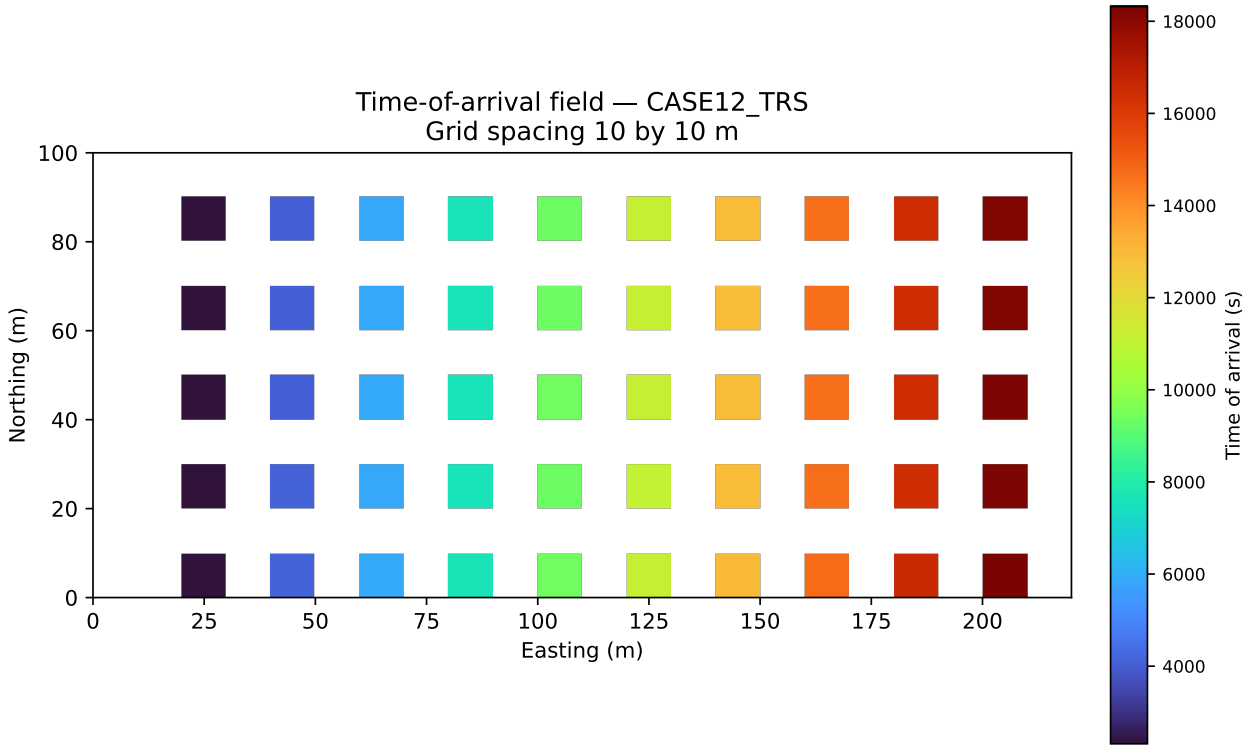


Figure 2: Whole-domain time-of-arrival field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

4 of 4 simulation conditions currently have complete outputs matching the generated inputs. The verification decision is **FAIL**.

Figure 3 presents four complementary temporal-resolution diagnostics. Panel (a) compares deposited load with the analytical integral defined in Section 2; panels (b) and (d) use CFL=0.5 as the numerical baseline; and panel (c) shows the accumulation history at the specified downwind structure.

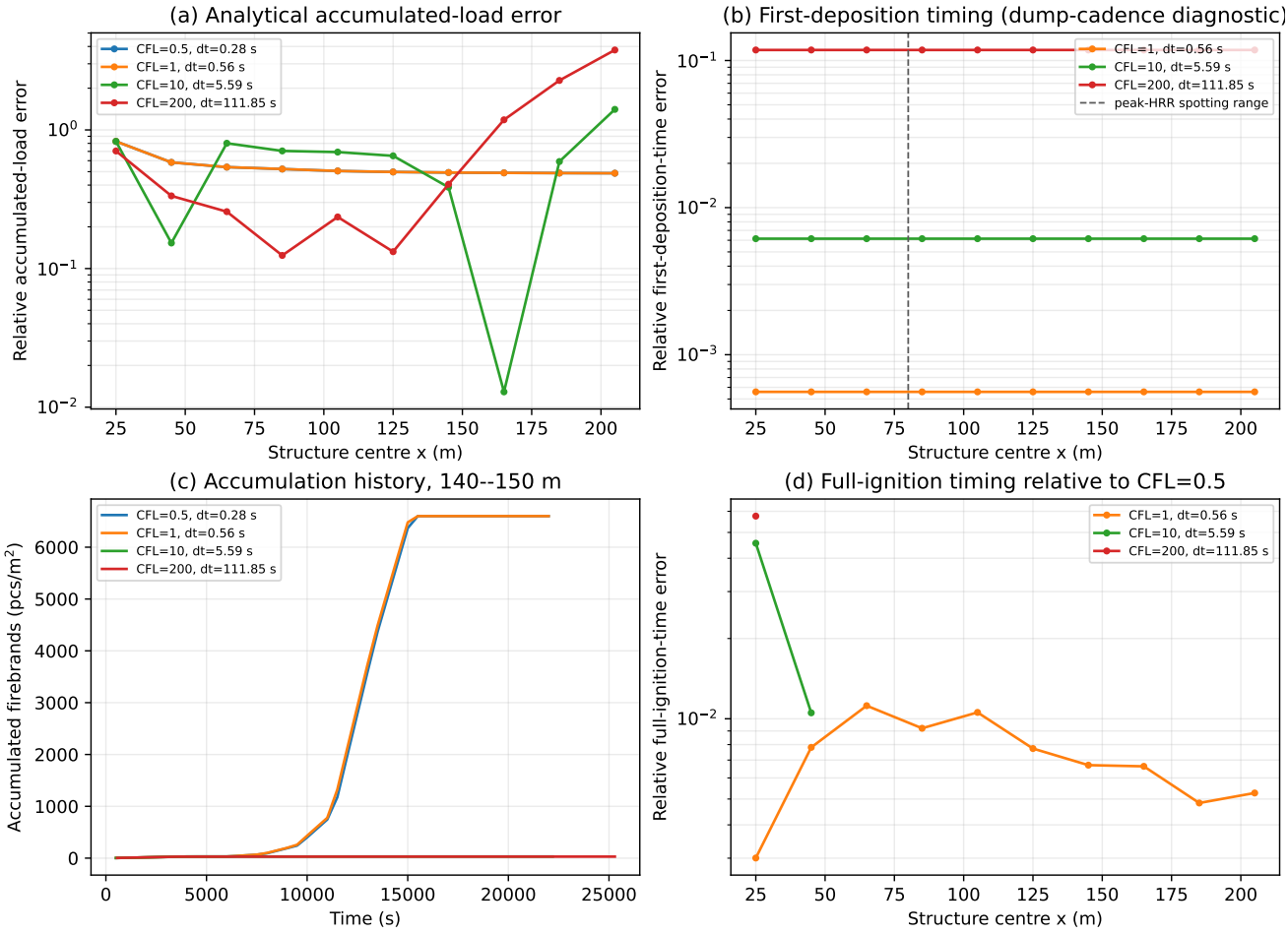


Figure 3: Temporal-resolution diagnostics for the prescribed temporal-resolution sweep: (a) accumulated-load relative error; (b) dump-cadence first-deposition-time relative error; (c) accumulation history at 140–150 m; and (d) full-ignition time relative error.

Figure 4 condenses the ignition comparison into mean relative error versus CFL and displays the acceptance threshold directly.

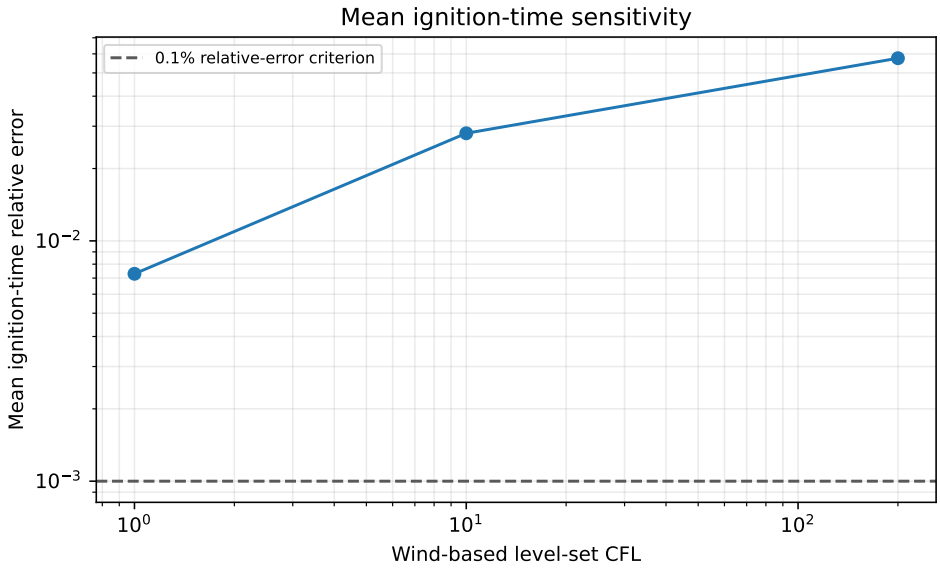


Figure 4: Mean full-ignition-time relative error referenced to CFL=0.5. The dashed line is the 0.1% acceptance level used for CFL=1 and 10.

7 Calculated metrics and verification decision

Table 3: Calculated verification metrics and component decisions.

Metric	Acceptance limit	Calculated	Status
Output completeness	4 simulation conditions	4/4	PASS
CFL 0.5 analytical-load mean error	≤ 0.01	0.54345	FAIL
CFL 1 analytical-load maximum error	≤ 0.001	0.82701	FAIL
CFL 10 analytical-load maximum error	≤ 0.001	1.407	FAIL
CFL 200 analytical-load maximum error	≤ 0.1	3.7738	FAIL
CFL 1 ignition-time mean relative error	≤ 0.001	0.0072949	FAIL
CFL 10 ignition-time mean relative error	≤ 0.001	0.028037	FAIL
CFL 1 accumulation-history NRMSE	≤ 0.05	0.0084659	PASS
CFL 10 accumulation-history NRMSE	≤ 0.05	0.62444	FAIL

The current decision is **FAIL**. A pass requires all four input-matched runs and all tabled acceptance checks. A not-run state is not evidence of agreement.

This is a deterministic code-verification test of the implemented idealized WUI model, not field validation. The analytical load uses simulated source ignition times so that it isolates generation and deposition from ignition chronology. First-deposition timing is limited by transient output cadence because the current executable does not write the internal firebrand arrival time array as a final raster; that diagnostic is therefore intentionally excluded from the decision specification.

8 Technical interpretation

With the declared building-only switch restored, all four current-source simulation conditions produce nonzero structural heat release, firebrand transport, input identity record-matched final rasters, and a

complete metric record. This invalidates the earlier zero-HRR and missing-output interpretations, both of which arose while diagnosing the stale executable or the temporary altered switch.

The temporal history is comparatively stable at moderate step sizes: the CFL 1 and CFL 10 accumulation-history NRMSE values are 0.008466 and 0.008443, both below 0.05. The stricter pointwise/reference components do not pass. The analytical-load errors are 0.54345 for CFL 0.5, 0.82701 for CFL 1, 0.82690 for CFL 10, and 0.83927 for CFL 200, all above their respective limits. The CFL 1 and CFL 10 mean ignition-time relative errors are 0.007295 and 0.01326, compared with the 0.001 limit. The overall result is therefore *FAIL*, not *NOT RUN* or *NOT EVALUABLE*.

Because every input identity record, final output, and integer timestep/stop relation is valid, these discrepancies are not caused by an admissible case-configuration error. They identify disagreement between the implemented temporal/structural spotting calculation and the predeclared analytical-load and ignition-time requirements.

9 Scope and limitations

The test varies only temporal resolution in a fixed academic WUI configuration. It verifies timestep sensitivity for the selected transport and ignition scales and does not establish spatial convergence or physical validation.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.